

Healthcare Care Transition Analytics: Development and Evaluation of an Interactive Data-Driven Framework for Monitoring the Unaccompanied Alien Children Care Transition Pipeline

Neha Malav
Independent Researcher, India
Corresponding author: Neha Malav

Abstract

Aims: This study aimed to develop and evaluate a reproducible, interactive analytics framework for examining operational transitions involving Unaccompanied Alien Children (UAC), with emphasis on apprehension, custody, transfer, ongoing care, and discharge activity.

Study Design: Retrospective descriptive and exploratory data-analytics study using an aggregate operational dataset.

Methodology: A Python-based analytical workflow was developed using Pandas, NumPy, Plotly, Matplotlib, and Streamlit. The workflow included data ingestion, date standardization, numeric normalization, missing-value handling, exploratory data analysis, temporal aggregation, ratio-based key performance indicators, comparative analysis, and interactive visualization. The resulting Streamlit application enables user-defined date filtering, KPI monitoring, monthly trend analysis, comparison of CBP custody with HHS care, care-pipeline visualization, filtered-data inspection, and CSV export.

Results: The resulting framework operationalizes five principal measures: children apprehended and placed in CBP custody, children in CBP custody, children transferred out of CBP custody, children in HHS care, and children discharged from HHS care. The implementation demonstrates substantial temporal variability in operational activity and provides a unified interface for examining inflow, intermediate custody, transfer, continuing care, and discharge. A transfer ratio is calculated as transfers divided by apprehensions, while a discharge ratio is calculated as discharges divided by children in HHS care. Because these measures represent flows and stocks from potentially different cohorts, the ratios are interpreted as operational throughput indicators rather than individual-level probabilities.

Conclusion: The developed framework demonstrates how reproducible data processing and interactive dashboards can convert aggregate administrative data into an accessible operational decision-support system. The architecture provides a foundation for future forecasting, anomaly detection, capacity modeling, geographic analytics, and real-time integration.

Keywords: healthcare analytics; care transitions; Unaccompanied Alien Children; operational analytics; healthcare dashboard; data visualization; Streamlit; Python; key performance indicators; decision support

1. Introduction

Care transitions represent periods in which individuals move between different stages, settings, providers, or organizations within a broader care system. Such transitions are operationally important because information, responsibility, resources, and individuals move simultaneously between system components. Poorly coordinated transitions can create fragmentation, delays, duplicated effort, and inefficient use of resources (Coleman & Boulton, 2003; Coleman et al., 2006).

The increasing availability of administrative and operational data provides an opportunity to monitor these processes quantitatively. However, the existence of data alone does not guarantee useful decision support. Large and heterogeneous datasets may contain inconsistent formats, missing values, multiple temporal scales, and measurements that represent fundamentally different concepts. Data structure, standardization, governance, integration, and interpretation remain important challenges in healthcare analytics (Kruse et al., 2016).

Healthcare dashboards have emerged as one approach for transforming operational data into information that can be interpreted by clinicians, administrators, analysts, and other stakeholders. Reviews of healthcare dashboards

have identified applications including key performance indicator monitoring, resource management, workflow optimization, tracking, public-health surveillance, and decision support (Rabiei & Almasi, 2022; Almasi et al., 2026).

The present study applies these principles to a care-transition setting involving Unaccompanied Alien Children (UAC). The U.S. Department of Health and Human Services describes the UAC program as a system involving children who lack a parent or legal guardian available to provide care in the United States. Operational oversight of this system therefore involves movement through multiple custody and care stages (U.S. Department of Health and Human Services, 2026).

The project reported in this paper, *Healthcare Care Transition Analytics*, was independently designed and developed as an end-to-end analytics application. The public repository describes the project as an analytical system focusing on CBP apprehensions, children in CBP custody, transfers from CBP to HHS, children in HHS care, and discharges from HHS care.

The central contribution of this work is the integration of these measures into a reproducible analytical pipeline and interactive dashboard rather than treating each variable as an isolated statistic.

2. Background and Related Work

2.1 Care-transition analytics

The concept of transitional care has traditionally been associated with maintaining continuity as individuals move between healthcare settings. Coleman and Boulton (2003) emphasized that individuals with complex care needs may be particularly vulnerable during transitions because care responsibilities and information must cross organizational boundaries.

Subsequent intervention research demonstrated that structured transition support can improve outcomes following movement between care settings. Coleman et al. (2006) evaluated a care-transition intervention incorporating a transition coach and patient-centered information tools and reported reductions in hospital readmission.

Although the population and operational setting of the present study differ from conventional hospital-to-home transitional care research, the underlying analytical principle is similar: transitions should be studied as processes rather than isolated events.

In the UAC setting, movement can be conceptualized as: **Apprehension** → **CBP Custody** → **Transfer** → **HHS Care** → **Discharge**. Each stage represents a different operational state or flow. Understanding the relationship between these stages can help identify periods requiring additional operational attention.

2.2 Healthcare data analytics

Healthcare systems increasingly generate large quantities of administrative and operational information. Big-data research has identified opportunities for quality improvement, population management, accessibility of information, and improved decision-making while simultaneously identifying challenges related to data structure, security, standardization, storage, transfer, and governance (Kruse et al., 2016).

The present project adopts a comparatively lightweight analytical architecture. Rather than requiring a large enterprise data warehouse, it demonstrates how a structured CSV dataset can be transformed into an interactive analytical environment using widely available open-source Python tools.

2.3 Healthcare dashboards

Dashboards provide a visual mechanism for summarizing large quantities of information and presenting key indicators in a form that supports monitoring and decision-making. A systematic review by Rabiei and Almasi (2022) identified functional dashboard requirements including reporting, customization, tracking, reminders, alerts, and performance assessment. A more recent umbrella review identified KPI monitoring, resource management, analysis and prediction, information integration, tracking, evidence-based decision-making, and public-health surveillance among common dashboard applications (Almasi et al., 2026).

The current application incorporates several of these principles through KPI cards, temporal charts, comparative visualization, filtering, and downloadable data.

3. Research Gap

Existing healthcare-dashboard literature demonstrates substantial use of dashboards for clinical, quality, and public-health monitoring. However, there remains value in developing domain-specific analytical frameworks that combine multiple operational stages of a care-transition pipeline.

The present project addresses this gap through three design choices. First, it integrates inflow, custody, transfer, continuing-care, and discharge variables into one analytical model. Second, it emphasizes temporal exploration, allowing stakeholders to investigate different date intervals rather than relying on a single static summary. Third, it explicitly distinguishes between flow variables and stock variables, an important consideration when constructing ratios from operational datasets.

Research question: Can a reproducible Python-based analytical workflow and interactive dashboard provide an interpretable framework for monitoring temporal patterns and transition-related operational indicators within the UAC care pipeline?

4. Objectives

- To preprocess and standardize an aggregate UAC operational dataset for analytical use.
- To characterize temporal variation in apprehensions, transfers, and discharges.
- To construct operational indicators for transfer and discharge activity.
- To visualize relationships between CBP custody, transfer activity, HHS care, and discharge.
- To develop an interactive dashboard that supports filtering, exploration, and export of analytical data.

5. Materials and Methods

5.1 Study design

This study used a retrospective descriptive and exploratory analytical design. The work did not involve prospective intervention, randomized allocation, individual clinical decision-making, or individual-level prediction. Instead, the analysis focused on aggregate operational observations contained in the project dataset.

5.2 Data source

The project repository contains a CSV dataset titled *HHS_Unaccompanied_Alien_Children_Program (1).csv*. The dataset contains dated observations for five principal operational measures: children apprehended and placed in CBP custody; children in CBP custody; children transferred out of CBP custody; children in HHS care; and children discharged from HHS care.

The project repository also contains the analytical notebook, application source code, requirements file, README documentation, and dashboard visualization assets.

5.3 Data preprocessing

Date conversion. The Date variable is converted into a Pandas datetime representation using error coercion. This enables chronological sorting, date filtering, monthly aggregation, and temporal visualization.

Numeric normalization. The operational variables may contain comma-separated numerical representations. The application removes commas before converting values to numeric form. For example, “2,484” is transformed to 2484.

Missing-value handling. Following numeric conversion, missing values in selected operational variables are replaced with zero. This permits subsequent aggregation and visualization without propagating missing numerical values through dashboard calculations. However, zero imputation should be interpreted carefully because a missing value does not necessarily mean that the true operational value was zero.

6. Analytical Framework

6.1 Care-transition pipeline

The analytical model conceptualizes the system as: **Apprehension** → **CBP Custody** → **Transfer** → **HHS Care** → **Discharge**. This model is useful for visualizing the operational sequence while recognizing that individuals may

remain within a stage for variable periods. The model should therefore not be interpreted as a same-day cohort flow.

6.2 Flow and stock variables

An important methodological distinction is made between flow and stock variables. Flow variables include apprehensions, transfers, and discharges. Stock variables include children in CBP custody and children in HHS care.

Conceptually: $\text{Stock}(t+1) = \text{Stock}(t) + \text{Inflow}(t) - \text{Outflow}(t)$. The exact relationship cannot be estimated from the aggregate variables alone because the available dataset does not provide complete individual-level cohort tracking. This distinction is essential when interpreting ratios.

7. Key Performance Indicators

7.1 Transfer ratio

$$\text{TR}(t) = \text{Transfers}(t) / \text{Apprehensions}(t)$$

The application calculates the ratio only when the apprehension denominator is non-zero. The measure should not be interpreted as the percentage of children apprehended on a particular day who were transferred on that same day. Transfers may include children apprehended during previous observation periods. Therefore, the measure is more appropriately described as a **transfer throughput ratio**.

7.2 Discharge ratio

$$\text{DR}(t) = \text{Discharges}(t) / \text{HHS Care}(t)$$

The ratio is calculated only when the HHS-care denominator is greater than zero. It should be treated as an operational throughput indicator rather than an individual probability of discharge.

8. Temporal Analysis

The application aggregates observations by calendar month. For each month, the system calculates totals for apprehensions, transfers, and discharges. Monthly aggregation reduces daily noise and provides a more interpretable view of medium-term operational variation.

The analytical notebook additionally includes rolling-average analysis of apprehension activity, supporting the distinction between short-term fluctuations and broader trends.

9. Dashboard Architecture

The system follows the architecture: **CSV Dataset** → **File Upload** → **Data Ingestion** → **Date Standardization** → **Numeric Cleaning** → **Missing-Value Handling** → **Date Filtering** → **KPI Calculation** → **Monthly Aggregation** → **Visualization** → **Filtered Dataset** → **CSV Export**.

The application is implemented in Streamlit and uses Pandas, NumPy, Plotly, and related Python functionality.

10. Interactive Dashboard Components

10.1 File upload

The dashboard begins with a CSV file-upload interface. This design allows the application to be reused with compatible future datasets rather than depending exclusively on one hard-coded file.

10.2 Date filtering

Users can select a start date and end date through the sidebar. The application then creates a filtered dataset containing observations within the selected period.

10.3 KPI cards

The dashboard displays six primary indicators: total apprehended, total transfers, average children in HHS care, total discharges, average transfer ratio, and average discharge ratio.

10.4 Monthly apprehension visualization

A Plotly line chart displays monthly apprehension totals, enabling users to identify increases, decreases, high-volume periods, low-volume periods, and changes over time.

10.5 Monthly transfer visualization

A second line chart displays monthly transfers. Examining this alongside apprehension activity can provide insight into whether downstream transfer activity changes in response to upstream demand.

10.6 Monthly discharge visualization

A third line chart displays monthly discharges and provides a downstream view of the care pipeline.

10.7 CBP versus HHS comparison

The dashboard calculates the average number of children in CBP custody and HHS care and displays the comparison using a bar chart.

10.8 Care-pipeline distribution

The application generates a donut-style visualization containing CBP custody, transferred, HHS care, and discharged measures. Because these quantities may represent different observation types and cohorts, the visualization should be interpreted as a distribution of reported operational measures rather than a literal cohort accounting diagram.

10.9 Filtered dataset and export

The dashboard displays the filtered dataset and provides a CSV download function, enabling users to move from visualization to further analysis without manually reproducing the filtering operation.

11. Results

11.1 Operational variation

The developed framework demonstrates that UAC care-transition activity can be examined simultaneously across multiple stages. The underlying project analysis identifies substantial fluctuation in CBP apprehension activity and temporal variation in apprehensions and transfers.

Rather than reducing the system to a single average, the dashboard enables users to examine the timing and magnitude of changes. This is important because a system operating under variable demand may experience substantially different resource requirements during high-volume periods compared with periods near its average activity level.

11.2 Transfer activity

The application calculates a transfer ratio for each observation in which apprehensions are greater than zero. The ratio provides a compact measure of the relationship between transfers and apprehensions. However, because transfers may involve children from earlier apprehension periods, a ratio greater than 1 should not be interpreted as evidence that more than 100% of newly apprehended children were transferred. The more defensible interpretation is the system's transfer throughput during an observation period relative to the contemporaneous apprehension flow.

11.3 Discharge activity

Discharge activity provides an indication of movement out of HHS care. The dashboard calculates the discharge ratio relative to the reported HHS-care population. Because HHS care represents a stock while discharge represents a flow, the ratio is best interpreted as a descriptive throughput indicator. A low ratio may indicate that discharge activity is small relative to the reported care population, but it cannot independently establish an individual's length of stay.

11.4 Temporal monitoring

The monthly visualizations provide a mechanism for identifying periods of changing operational demand. The project identifies substantial fluctuations in apprehensions and temporal variation in apprehension and transfer activity. The present study does not statistically establish seasonality; formal seasonal decomposition or inferential testing would be required before concluding that observed changes constitute statistically established seasonal effects.

12. Discussion

The principal contribution of this study is the development of an integrated analytical framework for examining a multi-stage care-transition pipeline. The project illustrates how a relatively simple operational dataset can be transformed into a decision-support interface using open-source technologies.

This is consistent with broader literature demonstrating that dashboards can support KPI monitoring, performance management, tracking, and decision-making (Rabiei & Almasi, 2022; Almasi et al., 2026).

The project also demonstrates an important methodological challenge: not all operational metrics can be interpreted as conventional percentages. A transfer-to-apprehension ratio may exceed 100% because the numerator and denominator can represent different cohorts. Similarly, a discharge-to-HHS-care ratio compares a flow with a stock. Therefore, operational analytics should not only calculate metrics but also communicate their semantic meaning.

13. Implications for Operational Decision Support

13.1 Demand monitoring

Increasing apprehension activity may indicate increasing upstream demand and the potential need for downstream capacity planning.

13.2 Transfer monitoring

Comparing apprehension and transfer trends may help identify periods in which transfer activity changes substantially relative to incoming activity.

13.3 Capacity monitoring

Monitoring the number of children in CBP custody and HHS care can provide a high-level indication of population accumulation.

13.4 Discharge monitoring

Discharge trends can provide information about downstream movement out of HHS care.

13.5 Period-specific investigation

Date filtering allows stakeholders to compare selected periods and investigate operational changes without modifying source code.

14. Strengths of the Framework

- **Reproducible implementation:** the project contains source code, documentation, analytical notebook, requirements, data, and visualization assets in a version-controlled repository.
- **End-to-end workflow:** the project extends beyond exploratory analysis to a functional interactive application.
- **Interactive analysis:** users can modify the observation window through date filters.
- **Transparent calculations:** KPI formulas are directly represented in the application source code, improving interpretability and reproducibility.
- **Exportability:** the filtered analytical dataset can be downloaded as a CSV file.
- **Extensibility:** the architecture can be extended with predictive analytics, automated reporting, geographic visualization, real-time integration, and advanced KPI monitoring.

15. Limitations

15.1 Aggregate data

The analysis is based on aggregate operational observations rather than longitudinal individual-level records. Therefore, the framework cannot estimate individual length of stay, individual probability of transfer, individual probability of discharge, individual clinical outcomes, or individual trajectories.

15.2 Flow-stock mismatch

The analysis compares flow and stock variables in several indicators. Future research should use cohort-based measures where individual-level data are available.

15.3 Transfer ratio

The transfer ratio is not equivalent to a same-day transfer probability. It should be interpreted as throughput relative to contemporaneous apprehension activity.

15.4 Discharge ratio

The discharge ratio cannot be used alone to infer average length of stay. Length-of-stay analysis would require longitudinal entry and exit dates.

15.5 Missing data and zero imputation

The application replaces missing numeric values with zero. Future versions should distinguish genuine zero values from missing observations.

15.6 No causal inference

The study is descriptive. Observed temporal relationships cannot establish causality.

15.7 No predictive model

The current implementation does not contain a validated forecasting or machine-learning model.

16. Ethical Considerations

The analysis uses aggregate operational data contained within the project repository and does not intentionally analyze directly identifiable individual-level information. Because the underlying population consists of children, privacy and responsible data governance are particularly important.

The dashboard is intended for aggregate operational monitoring and research purposes and should not be used as an autonomous mechanism for making individual decisions concerning care, custody, immigration status, eligibility, or other high-impact outcomes. Future versions using more granular data should incorporate appropriate privacy controls, access restrictions, data minimization, provenance documentation, and formal ethical review where required.

17. Practical Recommendations

- **Replace simple throughput ratios with cohort-aware measures.** Where sufficiently detailed data are available, calculate transfer completion for defined cohorts rather than same-day flow ratios.
- **Develop length-of-stay analytics.** Individual or cohort-level entry and discharge dates would enable direct estimation of median, mean, and percentile length of stay.
- **Develop predictive forecasting.** Future research should evaluate time-series models for forecasting apprehension volume, CBP custody, transfers, HHS-care population, and discharge volume.
- **Implement anomaly detection.** Automated alerts could identify unusually high or low observations using statistical process-control methods, rolling baselines, or machine-learning approaches.
- **Introduce capacity forecasting.** Forecasting models could be combined with operational capacity thresholds to estimate periods of potential pressure before they occur.
- **Expand geographic analysis.** Where authorized geographic data are available, the dashboard could incorporate regional and state-level analysis.
- **Improve data semantics.** Every KPI should clearly identify numerator, denominator, time interval, unit, population definition, flow/stock classification, and limitations.

18. Future Research

18.1 Cohort-based transition modeling

Future research could follow cohorts from apprehension through transfer and eventual discharge. A cohort model could be represented as: Cohort(t) → Transfer(t+k) → HHS Care(t+k) → Discharge(t+k+m), where k and m represent time intervals between stages.

18.2 Survival analysis

If individual-level data become available, survival analysis could estimate time to discharge and identify factors associated with prolonged stays.

18.3 Predictive analytics

The dashboard could evolve from descriptive monitoring to predictive decision support, including next-month demand forecasts, predicted care population, expected discharge volume, probability of capacity stress, and anomaly alerts.

18.4 Explainable artificial intelligence

If machine-learning models are incorporated, explainability should be prioritized so that operational stakeholders can understand the factors contributing to a forecast or alert.

18.5 Real-time data integration

The current CSV upload architecture could eventually be replaced or supplemented by automated data pipelines supporting ingestion, validation, transformation, analytics, dashboards, and alerts.

19. Conclusion

This study presented *Healthcare Care Transition Analytics*, an interactive, reproducible framework for monitoring a multi-stage UAC care-transition pipeline.

The system integrates apprehension, CBP custody, transfer, HHS care, and discharge measures into a unified analytical workflow. Using Python, Pandas, NumPy, Plotly, and Streamlit, the project transforms aggregate operational data into interactive KPIs, monthly trend visualizations, comparative analyses, pipeline visualization, filtered datasets, and downloadable outputs.

The main methodological contribution is the explicit treatment of the care-transition process as a dynamic pipeline rather than a collection of independent counts. The study also demonstrates why KPI semantics matter. A transfer-to-apprehension ratio can exceed 100% without implying that more than 100% of newly apprehended children were transferred, because the numerator and denominator can correspond to different cohorts. Similarly, a discharge-to-HHS-care ratio compares a flow with a stock and should therefore be interpreted as a descriptive throughput measure rather than an individual-level probability.

The resulting dashboard provides a practical foundation for operational monitoring while remaining transparent about the limitations of aggregate data. More advanced research should focus on cohort tracking, length-of-stay analysis, forecasting, anomaly detection, capacity modeling, geographic analytics, and real-time integration.

Overall, this work demonstrates how reproducible data analytics can bridge the gap between raw administrative data and interpretable operational decision support. The framework is technically extensible and provides a foundation for future research into predictive and intelligent care-transition analytics.

20. Data and Software Availability

The source code, project documentation, analytical notebook, dataset, and dashboard visualization assets are publicly available in the author's GitHub repository: *Healthcare Care Transition Analytics*. The repository contains the application source code, dataset, notebook, requirements, README documentation, and visualization assets.

Repository: <https://github.com/malavneha/Healthcare-Care-Transition-Analytics>

21. Authors' Contributions

Neha Malav: Conceptualization; study design; data acquisition; data preprocessing; exploratory data analysis; KPI design; software development; dashboard development; visualization; interpretation of results; documentation; manuscript preparation; revision; and final approval of the manuscript.

22. Competing Interests

The author declares that no competing interests exist.

23. Funding

No external funding was received for this study.

24. Consent and Ethical Approval

The study used aggregate operational data and did not involve direct interaction with human participants or analysis of identifiable individual-level records. Therefore, individual participant consent was not applicable to the present analysis. If future research uses identifiable or individual-level data, appropriate institutional ethical review and data-governance procedures should be obtained before analysis.

25. Acknowledgements

The author acknowledges the public availability of the operational dataset used in the project and the open-source Python ecosystem that enabled development of the analytical framework.

26. Generative Artificial Intelligence Disclosure

Generative artificial intelligence was used as an editorial and research-assistance tool during manuscript preparation, including assistance with organization, language refinement, literature-oriented drafting, and critical review of manuscript structure. The author independently developed the underlying software project and is responsible for the accuracy, interpretation, originality, and final content of the manuscript. All factual claims, calculations, references, and descriptions of the software implementation should be verified against the underlying dataset, source code, and cited literature before submission.

27. References

1. Almasi, S., Rabiei, R., Hamed, S. Z., Ghaderzadeh, M., & Bastani, P. (2026). Umbrella review of healthcare dashboards: Applications, benefits, design, and challenges. *DIGITAL HEALTH*, 12, 20552076261429609. doi:10.1177/20552076261429609
2. Coleman, E. A., & Boulton, C. (2003). Improving the quality of transitional care for persons with complex care needs. *Journal of the American Geriatrics Society*, 51(4), 556–557.
3. Coleman, E. A., Parry, C., Chalmers, S., & Min, S.-J. (2006). The care transitions intervention: Results of a randomized controlled trial. *Archives of Internal Medicine*, 166(17), 1822–1828.
4. Kruse, C. S., Goswami, R., Raval, Y. J., & Marawi, S. (2016). Challenges and opportunities of big data in health care: A systematic review. *JMIR Medical Informatics*, 4(4), e38.
5. Rabiei, R., & Almasi, S. (2022). Requirements and challenges of hospital dashboards: A systematic literature review. *BMC Medical Informatics and Decision Making*, 22, 287.
6. Xie, C. X., Chen, Q., Hincapié, C. A., Hofstetter, L., Maher, C. G., & Machado, G. C. (2022). Effectiveness of clinical dashboards as audit and feedback or clinical decision support tools on medication use and test ordering: A systematic review of randomized controlled trials. *Journal of the American Medical Informatics Association*, 29(10), 1773–1785.
7. Perdue, M. L. (2025). Creating interactive data dashboards for evidence syntheses. *Cochrane Evidence Synthesis and Methods*.
8. Malav, N. (2026). *Healthcare Care Transition Analytics* [Computer software]. GitHub.

28. Repository

Malav, N. (2026). *Healthcare Care Transition Analytics*. GitHub repository. <https://github.com/malavneha/Healthcare-Care-Transition-Analytics>